

Test_r

2025-10-29

```
library(tidyverse)
```

Introduction

The chosen dataset comes from tidytuesday and is a dataset regarding Pokemons. It can be found here: <https://github.com/rfordatascience/tidyuesday/blob/main/data/2025/2025-04-01>

Data presentation

A summary of the data set can be seen below. From this information, it is possible to see which variables makes sense to do statistical test's on.

As an example "id" and "species_id" does not make sense to use in these test's, because it is a index value more that data value.

Therefore for this assignment, the chosen variables for further statistical investigation are the Pokémons "attack" and "hp" data.

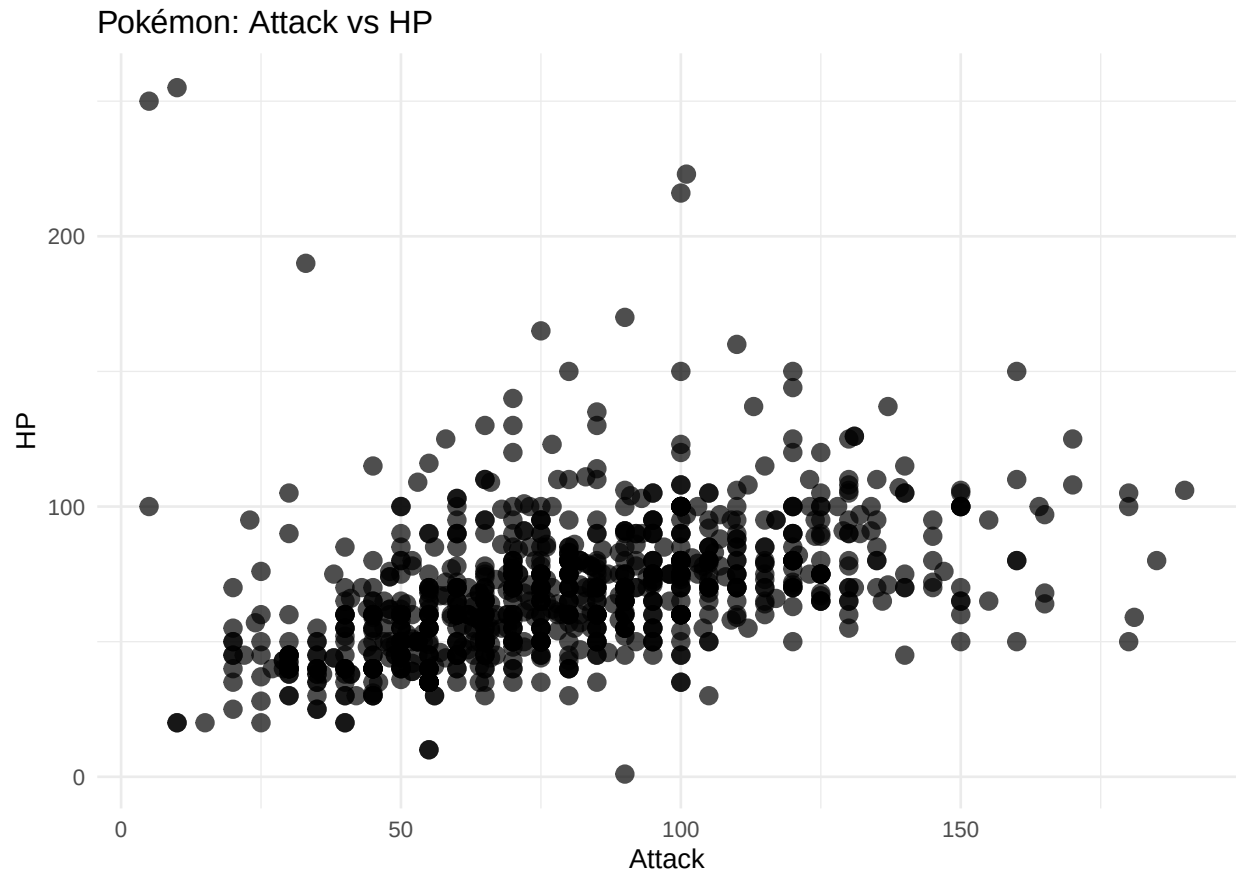
```
summary(pokemon_df)
```

```
##           id           pokemon           species_id           height
##  Min.      :    1  Length:949      Min.      :    1.0  Min.      : 0.100
## 1st Qu.:   238  Class :character 1st Qu.:191.0 1st Qu.: 0.500
## Median :   475  Mode  :character Median :395.0 Median : 1.000
## Mean    :  1900              Mean   :400.1 Mean   : 1.228
## 3rd Qu.:   712              3rd Qu.:615.0 3rd Qu.: 1.500
## Max.    :10147              Max.    :802.0 Max.    :14.500
##
##           weight  base_experience  type_1           type_2
##  Min.      : 0.10  Min.      : 36.0  Length:949  Length:949
## 1st Qu.:  8.50  1st Qu.: 68.0  Class :character  Class :character
## Median : 28.80  Median :157.0  Mode  :character  Mode  :character
## Mean    : 66.21  Mean   :150.5
## 3rd Qu.: 66.60  3rd Qu.:184.0
## Max.    :999.90  Max.    :608.0
##
##           hp           attack           defense  special_attack
##  Min.      : 1.00  Min.      : 5.00  Min.      : 5.00  Min.      :10.00
## 1st Qu.: 50.00  1st Qu.: 55.00  1st Qu.: 50.00  1st Qu.: 50.00
## Median : 65.00  Median : 75.00  Median : 70.00  Median : 65.00
## Mean    : 68.95  Mean    : 79.47  Mean    : 74.07  Mean    : 72.81
## 3rd Qu.: 80.00  3rd Qu.:100.00  3rd Qu.: 90.00  3rd Qu.: 95.00
## Max.    :255.00  Max.    :190.00  Max.    :230.00  Max.    :194.00
##
## special_defense  speed           color_1           color_2
```

```
## Min.    : 20.00   Min.    :  5.00   Length:949      Length:949
## 1st Qu.: 50.00   1st Qu.: 45.00   Class :character Class :character
## Median : 70.00   Median : 65.00   Mode  :character Mode  :character
## Mean   : 72.22   Mean    : 69.02
## 3rd Qu.: 90.00   3rd Qu.: 90.00
## Max.    :230.00   Max.     :180.00
##
##      color_f      egg_group_1      egg_group_2      url_icon
## Length:949      Length:949      Length:949      Length:949
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
## generation_id    url_image
## Min.    :1.000    Length:949
## 1st Qu.:2.000    Class :character
## Median :4.000    Mode  :character
## Mean    :3.695
## 3rd Qu.:5.000
## Max.    :7.000
## NA's    :147
```

To visualize the data, one can plot the two variables against each other. As seen below:

```
ggplot(pokemon_df, aes(x = attack , y = hp)) +
  geom_point(alpha = 0.7, size = 3) +
  theme_minimal() +
  labs(
    title = "Pokémon: Attack vs HP",
    x = "Attack",
    y = "HP"
  )
```



We setup a test, that tests if there is a correlation between the two variables.

Here our null hypothesis is that the more HP a Pokémon has the more attack it will have as well

```
cor_result <- cor.test(pokemon_df$attack, pokemon_df$hp)
```

```
cor_result$estimate # correlation coefficient
```

```
##          cor
## 0.4266479
```

```
cor_result$p.value # p-value
```

```
## [1] 2.904688e-43
```

From this we have a very low p-value, meaning that we will have to reject our null hypothesis. Furthermore if one were to fit a linear model to this data we can see the model fit very poorly, both visually as seen below, but it is also possible to examine the correlation coefficient of the fit.

```
library(ggplot2)
```

```
ggplot(pokemon_df, aes(x = hp, y = attack)) +
  geom_point(alpha = 0.6, color = "darkblue") + # scatter points
  geom_smooth(method = "lm", se = TRUE, color = "red") + # regression line
  theme_minimal() +
  labs(
    title = "Attack vs HP of Pokémon",
    subtitle = paste0("Correlation: ", round(cor_result$estimate, 2),
```

```

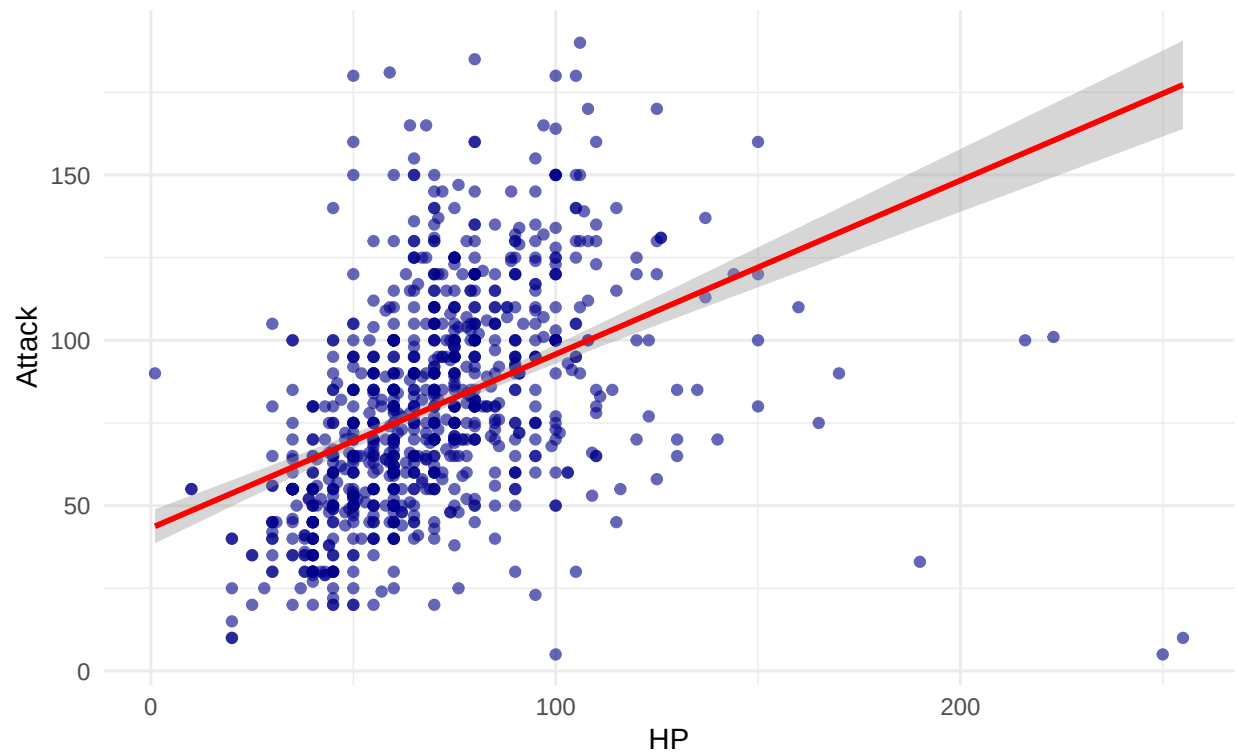
    ", p-value: ", signif(cor_result$p.value, 3)),
  x = "HP",
  y = "Attack"
)

```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Attack vs HP of Pokémon

Correlation: 0.43, p-value: 2.9e-43



'''